

Adonay Tesfamichael

IT-214-006

Soccer league

The soccer league company is trying to compare each soccer league year and save their information. With this diagram the league can keep track of every team in the clubs, throughout the year.

The main entities are:

- **SEASON:** all seasons in the league (name, registration start date, registration end date, season start date, season end date, year)
- **Surrogate PK:** ID_AT

attribute	Domain	Simple/Composite	Single/Multi	Required/Optional	Unique/Duplicated
ID_AT	INT	SIMPLE	SINGLE	REQUIRED	UNIQUE
NAME	VARCHAR (6)	SIMPLE	SINGLE	REQUIRED	DUPLICATED
REGISTRATION_START_DATE	DATE	SIMPLE	SINGLE	REQUIRED	DUPLICATED
REGISTRATION_END_DATE	DATE	SIMPLE	SINGLE	REQUIRED	DUPLICATED
SEASON_START_DATE	DATE	SIMPLE	SINGLE	REQUIRED	DUPLICATED
SEASON_END_DATE	DATE	SIMPLE	SINGLE	REQUIRED	DUPLICATED
YEAR	DECIMAL (4,0)	SIMPLE	SINGLE	REQUIRED	DUPLICATED

- **DIVISION:** all divisions in the league (age, sex, rank, email, phone)
- **Surrogate PK:** ID_AT
- **FK:** SEASON_AT_ID_AT references the pk ID_AT in the entity SEASON

attribute	Domain	Simple/Composite	Single/Multi	Required/Optional	Unique/Duplicated
ID_AT	INT	SIMPLE	SINGLE	REQUIRED	UNIQUE
AGE	VARCHAR (6)	SIMPLE	SINGLE	REQUIRED	DUPLICATED
SEX	VARCHAR (5)	SIMPLE	SINGLE	REQUIRED	DUPLICATED
RANK	INT	SIMPLE	SINGLE	REQUIRED	DUPLICATED
EMAIL	VARCHAR (25)	SIMPLE	SINGLE	REQUIRED	UNIQUE
PHONE	VARCHAR (12)	SIMPLE	SINGLE	REQUIRED	UNIQUE

- **CLUB:** all clubs in the league (name, email, phone, abbreviation)
- **Surrogate PK:** ID_AT

attribute	Domain	Simple/Composite	Single/Multi	Required/Optional	Unique/Duplicated
ID_AT	INT	SIMPLE	SINGLE	REQUIRED	UNIQUE
NAME	VARCHAR (20)	SIMPLE	SINGLE	REQUIRED	DUPLICATED
EMAIL	VARCHAR (25)	SIMPLE	SINGLE	REQUIRED	DUPLICATED
PHONE	VARCHAR (12)	SIMPLE	SINGLE	REQUIRED	DUPLICATED
ABBREVIATION	VARCHAR (5)	SIMPLE	SINGLE	OPTIONAL	DUPLICATED

- **TEAM:** all teams in the league (birth year, sex, name, email, phone)
- **Surrogate PK:** ID_AT
- **FK:** CLUB_AT_ID_AT references the pk ID_AT in the entity CLUB

attribute	Domain	Simple/Composite	Single/Multi	Required/Optional	Unique/Duplicated
ID_AT	INT	SIMPLE	SINGLE	REQUIRED	UNIQUE
BIRTH_YEAR	VARCHAR (6)	SIMPLE	SINGLE	REQUIRED	DUPLICATED
SEX	VARCHAR (5)	SIMPLE	SINGLE	REQUIRED	DUPLICATED
NAME	VARCHAR (21)	SIMPLE	SINGLE	REQUIRED	UNIQUE
EMAIL	VARCHAR (25)	SIMPLE	SINGLE	REQUIRED	DUPLICATED
PHONE	VARCHAR (12)	SIMPLE	SINGLE	REQUIRED	DUPLICATED

- **REGISTRAION:** all registrations in the league (registration date)
- **Surrogate PK:** ID
- **FK:** DIVISION_AT_ID_AT references the pk ID_AT in the entity DIVISON
- **FK:** TEAM_AT_ID_AT references the pk ID_AT in the entity TEAM

attribute	Domain	Simple/Composite	Single/Multi	Required/Optional	Unique/Duplicated
ID_AT	INT	SIMPLE	SINGLE	REQUIRED	UNIQUE
REGISTRAION_DATE	DATE	SIMPLE	SINGLE	REQUIRED	DUPLICATED

The main relationship are:

SEASON has DIVISIONS

- One to many relationships because in each season the league has several divisions.

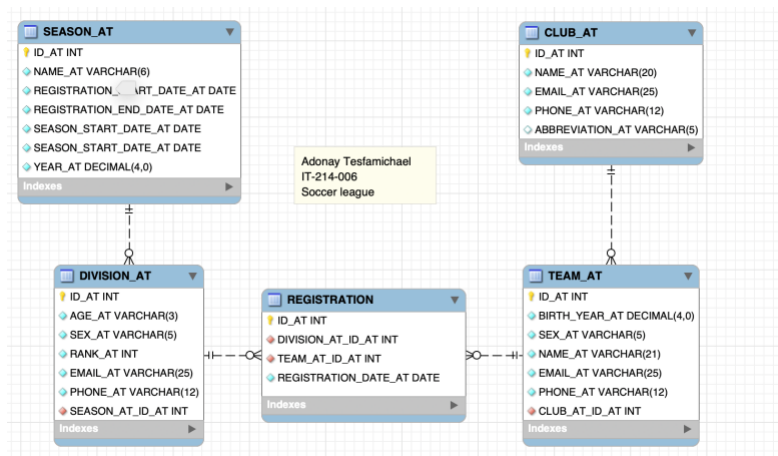
TEAM is assigned to CLUB

- One to many relationships because a club has MANY teams, and a team belongs to only ONE club.

TEAM is registered to DIVISION

- Many to many relationships because one team is assigned to a given division, while a division has many teams
- **BRIDGE:** REGISTRATION keeps the history of the team's membership.
- TEAM registers for division REGISTRATION
- DIVISION takes the teams REGISTRATION

ERD PICTURES



SEASON

Name: SEASON_AT					
Column	Datatype	PK	NN	UQ	
ID_AT	INT	☒	☒	☒	
NAME_AT	VARCHAR(6)	☐	☒	☐	
REGISTRATIO...	DATE	☐	☒	☐	
REGISTRATIO...	DATE	☐	☒	☐	
SEASON_ST...	DATE	☐	☒	☐	
SEASON_ST...	DATE	☐	☒	☐	
YEAR_AT	DECIMAL(4,0)	☐	☒	☐	

DIVISION

Name: DIVISION_AT					
Column	Datatype	PK	NN	UQ	
ID_AT	INT	☒	☒	☒	
AGE_AT	VARCHAR(3)	☐	☒	☐	
SEX_AT	VARCHAR(5)	☐	☒	☐	
RANK_AT	INT	☐	☒	☐	
EMAIL_AT	VARCHAR(25)	☐	☒	☒	
PHONE_AT	VARCHAR(12)	☐	☒	☒	
SEASON_AT_ID_AT	INT	☐	☒	☐	

CLUB

Name: CLUB_AT					
Column	Datatype	PK	NN	UQ	
ID_AT	INT	☒	☒	☒	
NAME_AT	VARCHAR(20)	☐	☒	☐	
EMAIL_AT	VARCHAR(25)	☐	☒	☐	
PHONE_AT	VARCHAR(12)	☐	☒	☐	
ABBREVIATION_AT	VARCHAR(5)	☐	☐	☐	

TEAM

Name: TEAM_AT					
Column	Datatype	PK	NN	UG	
ID_AT	INT	☒	☒	☐	
BIRTH_YEAR_AT	DECIMAL(4,0)	☐	☒	☐	
SEX_AT	VARCHAR(5)	☐	☒	☐	
NAME_AT	VARCHAR(21)	☐	☒	☒	
EMAIL_AT	VARCHAR(25)	☐	☒	☐	
PHONE_AT	VARCHAR(12)	☐	☒	☐	
CLUB_AT_ID_AT	INT	☐	☒	☐	

REGISTRATION

Name: REGISTRATION_AT					
Column	Datatype	PK	NN	UQ	
ID_AT	INT	☒	☒	☒	
DIVISION_AT_ID_AT	INT	☐	☒	☐	
TEAM_AT_ID_AT	INT	☐	☒	☐	
REGISTRATION_DATE_AT	DATE	☐	☒	☐	